Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 100201 | *Subject: Engineering Physics* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) For which angle of scattering is the Compton shift maximum
(i) 0^0 (iii) 180^0
(ii) 90^0 (iv) none of the above

(b) Which of the following is the characteristic of wave function?
(i) Continuous (iii) Normalizable
(ii) Single valued (iv) All of the above

(c) Critical angle depends on
(i) Refractive Index of rarer medium (iii) Refractive Index of both rarer and denser medium
(ii) Refractive Index of denser medium (iv) Refractive Index of none

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(d) Diamagnetic materials are
(i) Weakly repelled by magnetic field (iii) Weakly attracted by magnetic field
(ii) Strongly repelled by magnetic field (iv) Strongly attracted by magnetic field

(e) Write down time-independent Schrödinger's wave equation in 1-dimension.

(f) Young's double slit experiment was performed in a laboratory by taking monochromatic blue, orange and red light and fringe widths were obtained as β_B, β_O, β_R respectively. Other variables had been kept constants. Write down the fringe widths in decreasing order.

(g) Write down the Maxwell's equation which explains the non-existence of singular magnetic poles.

(h) What is Lorentz force?

(i) The maximum velocity and maximum acceleration in a simple harmonic oscillator are numerically equal. What is the time period?

(j) What is the ratio of intensities of maxima and minima in interference if the intensity ratio is 9 :1 ?

Q.2 (a) Mention the forces acting in Forced Harmonic Oscillator. Set its differential equation. Derive expressions for the phase difference and resultant amplitude. **[7]**

(b) What is Coriolis acceleration? Explain its applications in weather system. **[7]**

Q.3 (a) Discuss the structure and working of He-Ne LASER. Draw its energy band diagram. **[7]**

(b) What is population inversion? What is its importance in Lasing action? **[7]**

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Q.4 (a) Derive an expression for resultant intensity in Fraunhoffer's single slit diffraction. Mention the conditions for maxima and minima. Draw the intensities distribution curve. [7]

(b) Write a short note on Michelson's interferometer. [7]

Q.5 (a) Derive an expression for torque on a magnetic dipole inside a uniform magnetic field. Define magnetic moment. [7]

(b) What is ferromagnetic hysteresis? Draw the B-H plot and show the coercive field and remanent magnetisation in the plot. [7]

Q.6 (a) Discuss Kronig-Penney model for the motion of an electron in a periodic potential and the existence of allowed and forbidden energy bands. [7]

(b) Draw the total Energy (E) vs. Wave number (k) curve for an electron in 1-D lattice. [7]

Q.7 (a) What is Compton Effect? Give the expression for Compton shift. How does the shift depend upon the angle of scattering? [7]

(b) Derive Einstein's photo-electric equation. Plot the variation of frequency vs. stopping potential. Define threshold frequency. [7]

Q.8 Write down Schrödinger's wave equation for a particle inside 1-D Box given by [14]
$V(x) = 0, 0 < x < L = \infty$, $x < 0$ and $x > L$
Solve it to obtain energy eigen values. Show that energy eigen values are discrete.

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Amplitude Resonance
(b) Electronic Polarization in dielectrics
(c) 3-level LASER

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